

Appendix D

Metric	Calculation in the benchmark	Reason for selection
Request and failure counts	Count all measured requests and those whose recorded HTTP status was not 200.	Distinguish execution failures from successful requests that missed their target.
Relational correctness	Sum returned shard counts and compare with the expected count, or compare sorted returned movie identifiers with the sorted expected list. Count exact matches as passed checks.	Establish whether execution preserved the expected answer.
Client latency	Elapsed time from request encoding to HTTP response and JSON decoding. Vector generation was not included.	Measure latency experienced by the caller.
Vector retrieval time	From vector search to ranked candidates returned.	Identify the retrieval stage's contribution to latency.
Enrichment time	From relational enrichment, subsequent filtering and final truncation were complete.	Identify the enrichment stage's contribution to latency.
Total orchestration time	From entry into the orchestration method until construction of the response.	Distinguish service execution time from the broader client measurement.

Metric	Calculation in the benchmark	Reason for selection
Mean latency	Calculate the arithmetic mean of successful timing observations.	Summarise average execution cost.
P95 and P99 latency	Sort observations and select the value at one-based position	Describe slower requests that the mean can conceal. With 30 observations, P99 equals the maximum.
Hit Rate@10	count if the target is found within the first ten positions.	Measure how consistently the designated target was recovered.
Precision@10 — dense	Percentage of relevant movies returned in top 10 positions from the relevance set.	Report the planned precision measure against the designated relevance set.
Recall@10	Percentage of the relevant movies returned in top 10 from total number of relevant movies in the relevance set.	Measure recovery of the specified relevance set.
F1@10	Harmonic mean of precision and recall	Report the planned combined precision–recall measure.
Binary NDCG@10 - dense	The benchmark uses the binary-relevance version of the logarithmic formula presented by Manning et al. [5].	Reward recovery near the top of the ranking and penalises burying good results lower in the returned results.
Mean reciprocal rank — sparse	Each request received the inverse of the rank of the correct result, or 0 if the result was absent.	Measure how highly the known item was ranked.

Metric	Calculation in the benchmark	Reason for selection
Returned-result count	Count the final results after filtering and truncation.	Identify requests returning fewer than the requested ten results, which target recovery alone would not reveal.